

5. TECHNOLOGY REQUIREMENTS—MEMORY TECHNOLOGIES

CMOS logic and memory together form the predominant majority of semiconductor device production. The types of memory considered in this chapter are DRAM and non-volatile memory (NVM). The emphasis is on commodity, stand-alone chips, since those chips tend to drive the memory technology. However, embedded memory chips are expected to follow the same trends as the commodity memory chips, usually with some time lag. For both DRAM and NVM, detailed technology requirements and potential solutions are considered.

5.1. DRAM

For DRAM, the main goal is to continue scaling the footprint of the 1T-1C cell, to the practical limit of 4F2. The challenges are vertical transistor structures, high- κ dielectrics to improve the capacitance density, while keeping the leakage low. In general, technical requirements for DRAMs become more difficult with scaling. In the past several years, DRAM was introduced with many new technologies (e.g., 193 nm argon fluoride (ArF) immersion high-NA lithography with double patterning technology, improved cell FET technology including fin type transistor [22], buried word line/cell FET technology [23] and so on).

Since the DRAM storage capacitor gets physically smaller with scaling, the equivalent oxide thickness (EOT) must scale down sharply to maintain adequate storage capacitance. To scale the EOT, dielectric materials having high relative dielectric constant (κ) will be needed. Therefore, metal-insulator-metal (MIM) capacitors have been adopted using high- κ ($\text{ZrO}_2/\text{Al}_2\text{O}_3/\text{ZrO}_2$) [24] as the capacitor of DRAMs having the ground rules between 48nm and 30nm half-pitch. And this material evolution and improvement are continued until 20 nm HP and ultra high- κ (perovskite $\kappa > 50 \sim 100$) material are released. Also, the physical thickness of the high- κ insulator should be scaled down to fit the minimum feature size. Due to that, capacitor 3-D structure will be changed from cylinder to pillar shape.

On the other hand, with the scaling of peripheral CMOS devices, a low-temperature process flow is required for process steps after formation of these devices. This is a challenge for DRAM cell processes that are typically constructed after the CMOS devices are formed, and therefore are limited to low-temperature processing. The DRAM peripheral device requirement can relax Ioff but demands more Ion of low standby power (LSTP) devices. But, in the future, high- κ metal gate will be needed for sustaining the performance [25].

The other important topic is the migration from 6F2 to 4F2 cell. As the half-pitch scaling becomes very difficult, it is impossible to sustain the cost trend. The most promising way to keep the cost trend and increasing the total bit output by generation is changing the cell size factor (a) scaling (where $a = [\text{DRAM cell size}]/[\text{DRAM half pitch}]^2$). Currently 6F2 ($a = 6$) is the most common. For example, vertical cell transistor is required but still a couple of challenges are remaining. Other alternative is the use of 3D DRAM.

All in all, maintaining sufficient storage capacitance and adequate cell transistor performance are required to keep the retention time characteristic in the future. And their difficult requirements are increasing to continue the scaling of DRAM devices and to obtain the bigger product size (i.e., >16 Gb). In addition to that, if efficiency of cost scaling becomes poor in comparison with introducing the new technology, DRAM scaling will be stopped, and 3D cell stacking structure will be adopted, or a new DRAM concept will be adopted. 3D cell stacking and new concept DRAM are discussed but there is no clear path for further scaling beyond the 2D DRAM.

5.2. NVM—FLASH

There are several intersecting memory technologies that share one common trait—non-volatility. The requirements and challenges differ according to the applications, ranging from RFIDs that only require Kb of storage to high-density storage of hundreds of Gb in a chip. Nonvolatile memory may be divided into two large categories—Flash memories (NAND Flash and NOR Flash), and non-charge-based-storage memories. Nonvolatile memories are essentially ubiquitous, and a lot of applications use embedded memories that typically do not require leading edge technology nodes. The More Moore nonvolatile memory tables only track memory challenges and potential solutions for leading edge standalone parts.

Flash memories are based on simple one transistor (1T) cells, where a transistor serves both as the access (or cell selection) device and the storage node. At this time Flash memory serves more than 99% of applications.

When the number of stored electrons reaches statistical limits, even if devices can be further scaled and smaller cells achieved, the threshold voltage distribution of all devices in the memory array becomes uncontrollable and logic states

unpredictable. Thus memory density cannot be increased indefinitely by continued scaling of charge-based devices. However, effective density increase may continue by stacking memory layers vertically.

The economy of stacking by completing one device layer then another and so forth is questionable. As depicted in Figure MM-10 [26], the cost per bit starts to rise after stacking several layers of devices. Furthermore, the decrease in array efficiency due to increased interconnection and yield loss from complex processing may further reduce the cost-per-bit benefit of this type of 3D stacking. In 2007, a ‘punch and plug’ approach was proposed to fabricate the bit line string vertically to simplify the processing steps dramatically [26]. This approach makes 3D stacked devices in a few steps and not through repetitive processing, thus promised a new low-cost scaling path to NAND flash. Figure MM-10 illustrates one such approach. Originally coined bit-cost-scalable, or BiCS, this architecture turns the NAND string by 90 degrees from a horizontal position to vertical. The word line (WL) remains in the horizontal planes. As depicted in Figure MM-10, this type of 3D approach is much more economical than the stacking of complete devices, and the cost benefit does not saturate up to quite high number of layers.

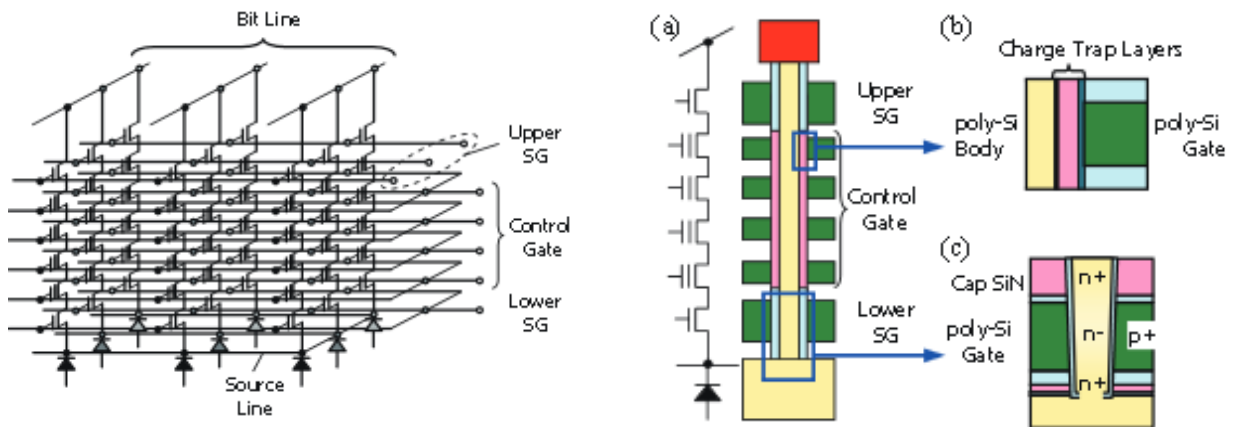


Figure MM-10 (left) A 3D NAND array based on a vertical channel architecture. (right) BiCS (bit cost scalable) – a 3D NAND structure using a punch and plug process [26].

A number of architectures based on the BiCS concept have been proposed since 2007 and several, including some that use floating gate instead of charge trapping layer for storage, have gone into volume production in the last 2–3 years. In general, all 3D NAND approaches have adopted a strategy of using much larger areal footprints than the conventional 2D NAND. The x- and y- dimensions (equivalent to cell size in 2D) of 3D NAND are in the range of 100nm and higher compared to ~15nm for the smallest 2D NAND. The much larger “cell size” is made up by stacking a large number of memory layers to achieve competitive packing density.

The economics of 3D NAND is further confounded by its complex and unique manufacturing needs. Although the larger cell size seems to relax the requirement for fine line lithography, to achieve high data rate it is desirable to use large page size and this in turn translates to fine pitched bit lines and metal lines. Therefore, even though the cell size is large metal lines still require ~20nm half-pitch that is only achievable by 193i lithography with double patterning. Etching of deep holes is difficult and slow, and the etching throughput is generally very low. Depositing many layers of dielectric and/or polysilicon, as well as metrology for multilayer films and deep holes all challenge unfamiliar territories. These all translate to large investment in new equipment and floor space and new challenges for wafer flow and yield.

The ultimate unknown is how many layers can be stacked. There seems no hard physics limit on the stacking of layers. Beyond certain aspect ratio (100:1 perhaps?) the etch-stop phenomenon, when ions in the reactive ion etching process are bent by electrostatic charge on the sidewall and cannot travel further down, may limit how many layers can be etched in one operation. However, this may be bypassed by stacking fewer layers, etching, and stacking more layers (at higher cost). Stacking many layers may produce high stress that bends the wafer and although this needs to be carefully engineered it does not seem to be an unsolvable physics limit. Even at 200 layers (at ~50nm for each layer) the total stack height is about 10μm, which is still in the same range as 10–15 metal layers for logic IC’s. This kind of layer thickness does not significantly affect bare die thickness (thinnest is about 40μm so far) yet. However, at 1000 layers the total layer thickness may cause thick dies that do not conform to the form factor for stacking multiple dies (e.g., 16 or 32) in a thin package. At this time, >250 layers are in volume production and there is optimism that 500+ layers are achievable and even 1000 layers are possible. In addition to processing challenges stacking more layers also increases the area overhead required to making

contacts to more word lines. This area overhead, plus the added processing complexity, will eventually diminish the cost benefit by adding more layers.

Renewed shrinking of the areal x-y footprint may eventually start when stacking more layers proves to be too difficult. However, such a trend is not guaranteed. If the hole aspect ratio is the limitation, shrinking the footprint would not reduce the ratio and would thus not be helpful. Furthermore, the larger cell size seems to at least partially contribute to the better performance of 3D NAND (speed and cycling reliability) compared to tight-pitch 2D NAND. Whether x-y scaling can still deliver such performance is not clear. Therefore, the roadmap projections for future generations stay the same as the current node in 2024. On the other hand, increasing the number of storage bits per memory cell, although technically challenging, seems to make progress. This is partly to take advantage that 3D NAND devices are intrinsically larger thus with more stored electrons and easier to make into more logic levels. At this time 4-bits/cell devices (QLC) are in volume production, and there is optimism that 5-bits/cell and more may become viable in the near future. Higher number of storage bits in a cell requires some trade off in performance since it takes longer to program and read, and reliability suffers when squeezing logic levels close together. Yet for many read-intensive applications such trade-offs are acceptable for lower cost.

5.3. NVM—EMERGING

Since 2D NAND Flash scaling is limited by statistical fluctuation due to too few stored charges, several non-conventional non-volatile memories that are not based on charge storage, so-called resistive or capacitive memories (ferroelectric or FeRAM, magnetic or MRAM, phase-change or PCRAM, and resistive or ReRAM), are being developed and form the category often called “emerging” memories. Even though 2D NAND is being replaced by 3D NAND (that is no longer subject to the drawback of too few electrons) some characteristics of non-charge based emerging memories, such as low voltage operation, or random access, are attractive for various applications and thus continue to be developed. These emerging memories usually have a two-terminal structure (e.g., resistor or capacitor) thus are difficult to also serve as the cell-selection device. The memory cell generally combines a separate access device in the form of 1T-1C, 1T-1R, or 1D-1R.

5.3.1. FeRAM

FeRAM devices achieve non-volatility by switching and sensing the polarization state of a ferroelectric capacitor. To read the memory state the hysteresis loop of the ferroelectric capacitor must be traced and the stored datum is destroyed and must be written back after reading (destructive read, like DRAM). Because of this ‘destructive read’ it is a challenge to find ferroelectric and electrode materials that provide both adequate change in polarization and the necessary stability over extended operating cycles. Many ferroelectric materials are foreign to the normal complement of CMOS fabrication materials, and can be degraded by conventional CMOS processing conditions. FeRAM is fast, low power, and low voltage and thus is suitable for RFID, smart card, ID card, and other embedded applications. Processing difficulty limits its wider adoption. Recently, HfO₂ based ferroelectric FET, for which the ferroelectricity serves to change the V_t of the FET and thus can form a 1T cell similar to Flash memory, has been proposed. If developed to maturity this new memory may serve as a low power and very fast Flash-like memory.

5.3.2. MRAM

Magnetic RAM (MRAM) devices employ a magnetic tunnel junction (MTJ) as the memory element. An MTJ cell consists of two ferromagnetic materials separated by a thin insulating layer that acts as a tunnel barrier. When the magnetic moment of one layer is switched to align with the other layer (or to oppose the direction of the other layer) the effective resistance to current flow through the MTJ changes. The magnitude of the tunneling current can be read to indicate whether a ONE or a ZERO is stored. Field switching MRAM probably is the closest to an ideal “universal memory” since it is non-volatile and fast and can be cycled indefinitely. Thus, it may be used as NVM as well as SRAM and DRAM. However, producing a magnetic field in an IC circuit is both difficult and inefficient. Nevertheless, field switching MTJ MRAM has successfully been made into products. The required magnetic field for switching, however, increases when the storage element scales while electromigration limits the current density that can be used to produce higher H field. Therefore, it is expected that field switch MTJ MRAM is unlikely to scale beyond 65nm node. Recent advances in “spin-transfer torque (STT)” approach where a spin-polarized current transfers its angular momentum to the free magnetic layer and thus reverses its polarity without resorting to an external magnetic field offer a new potential solution. During the spin transfer process, substantial current passes through the MTJ tunnel layer and this stress may reduce the writing endurance. Upon further scaling the stability of the storage element is subject to thermal noise, thus perpendicular magnetization materials are projected to be needed at 32nm and below. Perpendicular magnetization has been recently demonstrated.

With rapid progress of NAND Flash and the recent introduction of 3D NAND that promises to continue the equivalent scaling, the hope of STT-MRAM to replace NAND seems remote. However, its SRAM-like performance and much smaller footprint than the conventional 6T-SRAM have gained much interest in that application, especially in mobile devices that

do not require high cycling endurance, as in computation. Therefore, STT-MRAM is now mostly considered not as a standalone memory but an embedded memory [27], and is not tracked in the standalone NVM table. STT-MRAM would be a potential solution not only for embedded SRAM replacement but also for embedded Flash (NOR) replacement. This may be particularly interesting for IoT applications since low power is the most important. On the other hand, for other embedded systems applications using higher memory density, NOR Flash is expected to continue to dominate since it is still substantially more cost effective. Furthermore, Flash memory is well established for being able to endure the PCB board soldering process (at $\sim 250^{\circ}\text{C}$) without losing its preloaded code, which many emerging memories have not been able to demonstrate yet.

5.3.3. PCRAM and Crosspoint Memory

PCRAM devices use the resistivity difference between the amorphous and the crystalline states of chalcogenide glass (the most commonly used compound is $\text{Ge}_2\text{Sb}_2\text{Te}_5$, or GST) to store the logic levels. The device consists of a top electrode, the chalcogenide phase change layer, and a bottom electrode. The leakage path is cut off by an access transistor (or diode) in series with the phase change element. The phase change write operation consists of: (1) RESET, for which the chalcogenide glass is momentarily melted by a short electric pulse and then quickly quenched into amorphous solid with high resistivity, and (2) SET, in which a lower amplitude but longer pulse (usually $>100\text{ns}$) anneals the amorphous phase into low resistance crystalline state. The 1T-1R (or 1D-1R) cell is larger or smaller than NOR Flash, depending on whether MOSFET or BJT (or diode) is used. The device may be programmed to any final state without erasing the previous state, thus providing substantially faster programming throughput. The simple resistor structure and the low voltage operation also make PCRAM attractive for embedded NVM applications. The major challenges for PCRAM are the high current (fraction of mA) required to reset the phase change element, and the relatively long set time and high temperature tolerance to retain the preloaded code during solder reflow (at $\sim 250^{\circ}\text{C}$). Thermal disturb is a potential challenge for the scalability of PCRAM. However, thermal disturb effect is non-cumulative (unlike Flash memory in which the program and read disturbs that cause charge injection are cumulative) and the higher temperature RESET pulse is short (10ns). Interaction of phase change material with electrodes may pose long-term reliability issues and limit the cycling endurance and is a major challenge for DRAM-like applications. Like DRAM, PCRAM is a true random access, bit alterable memory.

The scalability of PCRAM device to $< 5\text{nm}$ has been demonstrated using carbon nanotubes as electrodes [28], and the reset current followed the extrapolation line from larger devices. In at least one case, cycling endurance of $1\text{E}11$ was demonstrated [29]. Phase change memory has been used in feature phones to replace NOR Flash since 2011, and has been in volume production at $\sim 45\text{nm}$ node since 2012, but no new product has been introduced since then. PCM memories have been also targeted in the last years as potential candidates for eFlash replacement for embedded applications [30]. In these works alloying of phase change materials of different classes allowed to obtain memory compliant to soldering reflow; however, such high temperature stability has come at the expense of slower write speed.

A 3D cross point memory (3D XP) has been reported [31]. Details are still lacking but it is speculated that the threshold switching (OTS) property of chalcogenide-based phase change material constitutes the core of the selector device responsible for the cross point cell, which was first reported in 2009 [32]. This is the first commercial realization of the widely published storage class memory (SCM) [33]. Computer systems badly need improved I/O throughput and reduce power and cost, and it is a promising candidate to change the entire memory hierarchy not only for high-end computation but for mobile systems as well. In addition, since the memory including the selector device is completely fabricated in the BEOL process it is relatively inexpensive to stack multiple layers to reduce bit cost.

3D cross point memory (3D XP) consists of a selector element made of ovonic threshold switching (OTS) (or an equivalent device) in series with a storage element. The selector device has a high ON/OFF ratio and is at OFF state at all times except when briefly turned on during writing or reading. The storage element is programmed to various logic states. Since the selector is always off, with high resistance the memory array has no leakage issue even if all storage elements are at low resistance state. During write or read operation the selector is temporarily turned on (by applying a voltage higher than its threshold voltage) and the OTS characteristic suddenly reduces its resistance to a very low value, allowing reading (or programming) current to be dominated by the resistance of the storage element. The storage element may be a phase-change material and in that case the memory cell is a phase-change RAM (PCRAM) switched by OTS. The storage element may also be a resistive memory material. Although bipolar operation makes the circuitry and operation more complicated, the array structure is very similar to that using PCRAM.

3D XP memory has entered commercial application for several years now, providing SCM function for servers and at some time also for PCs. Up to this date, however, it is not widely adopted since existing products use a proprietary interface. Yet, its unique advantage of random access and low cost making it a good choice for augmenting DRAM function in a memory

system thus its potential for wider adoption still exists. Its eventual success will depend on whether it can continue to hold a cost advantage over DRAM.

However, in the past two years the 3D XP memory has been withdrawn from the market and is currently not available. The exact reason is not announced. The commercial failure may be partially caused by the lack of common standard interface, and partly due to advances in DRAM scaling which narrows the cost differential.

Instead, recent attentions are steered toward a similar “selector only memory” (SOM). According to recent publications, SOM operates on a bi-directional mode, and uses only the selector but not the phase-change memory element. The selector serves both as the isolation device (like in 3D XP array) and as the storage device. The resistance of the selector changes by the direction of the applied voltage. SOM is still in research mode currently. It is not clear when a commercially available SCM based on this approach will be introduced.

Note that SLC 3D NAND can also serve SCM function with the proper interface, at much lower cost. But its considerably slower latency makes it more suitable for storage SCM, and 3D XP is more suitable for memory-type SCM (i.e., more DRAM-like).

5.3.4. *Resistive Memory (ReRAM)*

A large category of two-terminal devices, in which memory state is determined by resistivity of a metal-insulator-metal (MIM) structure, are being studied for memory applications. Many of these resistive memories are still in research stage and are discussed in more detail in the Beyond CMOS roadmap chapter. Because of their promise to scale below 10nm, and operate at extremely high frequencies ($< \text{ns}$) with low power consumption, the focused R&D efforts in many industrial labs in the last decade make this technology widely considered a potential successor to NAND (including 3D NAND). Being a two-terminal device, high-density ReRAM development has been limited by the lack of a good selector device. Recent advances in 3D XP memory, however, seem to have solved this bottleneck and ReRAM could make rapid progress if other technical issues such as erratic bits are solved. In addition to 3D XP array (like PCRAM-based 3D XP memory) high-density ReRAM products may be fabricated using a 2D array and small word-line (WL), and small bit-line (BL) half pitch. Furthermore, if eventually the OTS type of selector device is adopted it seems feasible to fabricate BiCS type 3D ReRAM using a transistor in the bottom and OTS selector for each ReRAM device in the 3D array, as depicted in Figure MM-11 [34]. Although no high-density ReRAM product has been introduced yet since the bottleneck of bipolar selector devices seems solved by the introduction of 3D XP memory, progress in ReRAM may be reasonably expected. Recently, however, the passion for developing high-density ReRAM seems to have dissipated. This may be due to two reasons. (1) the success of 3D NAND Flash has increased the entrance barrier and (2) difficulty in meeting the reliability requirements for large arrays. (Note that several announcements were made for successful development of ReRAM for smaller, Mb-size, arrays for embedded applications.)

In the last several years these above issues seem to doom large scale application of high-density ReRAM. The original argument that ReRAM, because it consists of thousands of atoms, is free from statistical fluctuation seems questionable now. It seems the filament that operates the ReRAM is made up by only a few atoms (ions). There seems evidence that even relatively large ReRAM device is subject to statistical fluctuation. Therefore, we stop tracking ReRAM for high-density application in this roadmap release.

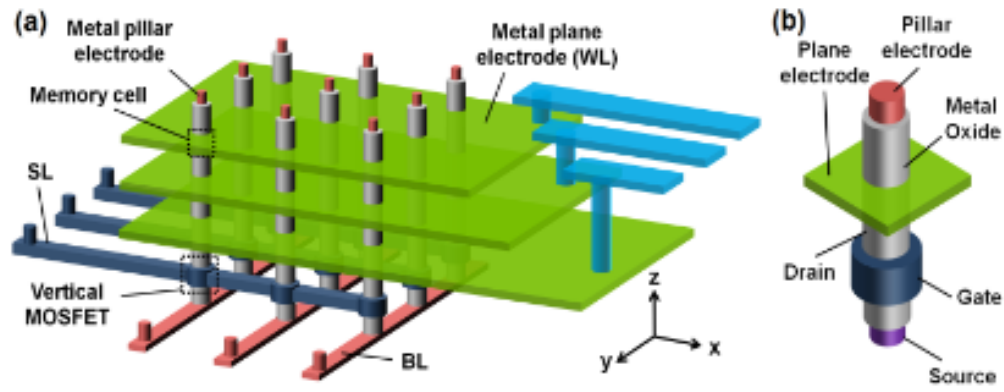


Figure MM-11 Schematic view of (a) 3D cross-point architecture using a vertical RRAM cell and (b) a vertical MOSFET transistor as the bit-line selector to enable the random access capability of individual cells in the array [34].